

1 Solution — GIAM 1.5

Problem 2

1.1 Problem

Find the **gcd**'s and **lcm**'s of the following pairs of numbers.

a	b	$\gcd(a, b)$	$\text{lcm}(a, b)$
110	273		
105	42		
168	189		

Table 1.1

1.2 Answer

a	b	$\gcd(a, b)$	$\text{lcm}(a, b)$
110	273	1	30030
105	42	21	210
168	189	21	1512

Table 1.2

1.3 Method — Prime Factorizations

Write each number as a product of primes. Then for every prime p :

- **gcd** takes the **smaller** exponent of p (the most that divides *both*);
- **lcm** takes the **larger** exponent of p (the least that *both* divide into).

Pair 1 — 110 and 273.

$$110 = 2 \cdot 5 \cdot 11, \quad 273 = 3 \cdot 7 \cdot 13.$$

They share *no* prime, so they are **coprime**: $\text{gcd} = 1$. With no overlap the **lcm** is the full product:

$$\text{lcm} = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 = 110 \cdot 273 = 30030.$$

Pair 2 — 105 and 42.

$$105 = 3 \cdot 5 \cdot 7, \quad 42 = 2 \cdot 3 \cdot 7.$$

Shared primes are **3** and **7**, so $\text{gcd} = 3 \cdot 7 = 21$. Taking the larger power of each prime (**2, 3, 5, 7**):

$$\text{lcm} = 2 \cdot 3 \cdot 5 \cdot 7 = 210.$$

Pair 3 — 168 and 189.

$$168 = 2^3 \cdot 3 \cdot 7, \quad 189 = 3^3 \cdot 7.$$

Shared primes are **3** (smaller power 3^1) and **7**, so $\text{gcd} = 3 \cdot 7 = 21$. The larger power of each prime gives

$$\text{lcm} = 2^3 \cdot 3^3 \cdot 7 = 8 \cdot 27 \cdot 7 = 1512.$$

1.4 Both Methods Side by Side

Two methods. Left: prime factorizations — gcd takes the LOWER power of each shared prime, lcm the HIGHER.
 Right: the Euclidean algorithm — repeated remainders; the last nonzero remainder is the gcd.

Pair 1: a = 110, b = 273 110 = 2 · 5 · 11 273 = 3 · 7 · 13 <i>no shared primes → coprime</i> gcd = 1, lcm = 2·3·5·7·11·13 = 30030	273 = 2·110 + 53 110 = 2·53 + 4 53 = 13·4 + 1 4 = 4·1 + 0 ← stop last nonzero remainder = gcd = 1
Pair 2: a = 105, b = 42 105 = 3 · 5 · 7 42 = 2 · 3 · 7 <i>shared: 3 and 7</i> gcd = 3·7 = 21, lcm = 2·3·5·7 = 210	105 = 2·42 + 21 42 = 2·21 + 0 ← stop last nonzero remainder = gcd = 21
Pair 3: a = 168, b = 189 168 = 2³ · 3 · 7 189 = 3³ · 7 <i>shared: 3 (lower power 3¹) and 7</i> gcd = 3·7 = 21, lcm = 2³·3³·7 = 1512	189 = 1·168 + 21 168 = 8·21 + 0 ← stop last nonzero remainder = gcd = 21

Cross-check every row: $\gcd(a,b) \cdot \text{lcm}(a,b) = a \cdot b$. (1·30030 = 110·273; 21·210 = 105·42; 21·1512 = 168·189.)

Figure 1.1: For each pair: the left column factors a and b into primes — gcd multiplies the lower power of each shared prime, lcm the higher. The right column runs the Euclidean algorithm, taking remainders until one is zero; the last nonzero remainder is the gcd. Every row satisfies $\gcd \cdot \text{lcm} = a \cdot b$.

1.5 Method — The Euclidean Algorithm

The Section 1.5 way to find a gcd without factoring is **Euclid's algorithm**: replace (a, b) by $(b, a \bmod b)$ repeatedly; the last nonzero remainder is the gcd.

Pair 1: $273 = 2 \cdot 110 + 53$, $110 = 2 \cdot 53 + 4$, $53 = 13 \cdot 4 + 1$, $4 = 4 \cdot 1 + 0 \Rightarrow \gcd = 1$.

Pair 2: $105 = 2 \cdot 42 + 21$, $42 = 2 \cdot 21 + 0 \Rightarrow \gcd = 21$.

Pair 3: $189 = 1 \cdot 168 + 21$, $168 = 8 \cdot 21 + 0 \Rightarrow \gcd = 21$.

Once the gcd is known, the lcm follows instantly from the identity in the next remark — no factoring needed.

1.6 Remark — The Identity $\gcd(a, b) \cdot \text{lcm}(a, b) = a b$

For any positive integers, the product of the **gcd** and the **lcm** equals the product of the numbers:

$$\text{gcd}(a, b) \cdot \text{lcm}(a, b) = a b.$$

This is exactly the “min exponent + max exponent = both exponents” fact, prime by prime: for each p , $\min(e_a, e_b) + \max(e_a, e_b) = e_a + e_b$. It also gives the fastest way to get each **lcm** above — divide the product by the **gcd**:

pair	$\frac{a b}{\text{gcd}} = \text{lcm}$	check
110, 273	$\frac{110 \cdot 273}{1}$	= 30030 ✓
105, 42	$\frac{105 \cdot 42}{21} = \frac{4410}{21}$	= 210 ✓
168, 189	$\frac{168 \cdot 189}{21} = \frac{31752}{21}$	= 1512 ✓

Table 1.3

1.7 Remark — Coprime Pairs and the Primorial 30030

Pair 1 is **coprime** ($\text{gcd} = 1$): 110 and 273 share no prime factor even though neither is prime. For coprime numbers the **lcm** is just the product, which is why $\text{lcm}(110, 273) = 30030$ is so large relative to the inputs. That 30030 is no accident: it is $2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13$, the product of the first six primes — the *primorial* $13\#$. Between them, $110 = 2 \cdot 5 \cdot 11$ and $273 = 3 \cdot 7 \cdot 13$ happen to use each of those six primes exactly once.

1.8 Remark — Why Min-for-gcd and Max-for-lcm

The rule is forced by what divisibility *means* in terms of exponents. A number $d = \prod p^{f_p}$ divides $a = \prod p^{e_p}$ exactly when $f_p \leq e_p$ for every prime. So:

- a *common divisor* of a, b needs $f_p \leq e_a$ and $f_p \leq e_b$, i.e. $f_p \leq \min(e_a, e_b)$ — the **greatest** such d takes $f_p = \min$;

- a *common multiple* m needs $a \mid m$ and $b \mid m$, i.e. exponent $\geq \max(e_a, e_b)$ — the **least** such m takes exponent = **max**.

This is the same min/max bookkeeping seen with exponents throughout the chapter, and it makes the **gcd-lcm** pair a clean illustration of unique factorization at work.

1.9 Remark — Factoring vs. Euclid

Both methods give identical answers here, but they scale very differently. The prime-factorization method is wonderfully transparent for small numbers like these, but it secretly requires *factoring* — which becomes infeasible for large inputs (the hardness of factoring is what RSA encryption rests on). Euclid’s algorithm, by contrast, never factors anything: it only takes remainders, and it finishes in a number of steps roughly proportional to the number of *digits* of the inputs. For Pair 1, factoring and Euclid each took about four lines; for two thousand-digit numbers, factoring is hopeless while Euclid still finishes almost instantly. That is why Euclid’s algorithm — not factorization — is the one actually used to compute **gcd**’s in practice.